# SRI CHAITANYA ACADEMY

RAMANBHAVAN-1,VSP

**MATHEMATICS**

JSN SIR DATE : 5/09/2023

1. The differential equation $\left(xy^3(1+\cos x)-y\right)dx+x\,dy=0$ represents the curve $\frac{x^2}{2y^2}=\frac{x^3}{b}+x^2\sin x+cx\cos x-d\sin x+k$ then $b+c+d$ is equal to $(k\in R)$

A) 6 B) 7 C) 10 D) 9

2. If $y=f(x)$ is solution of differential equation $\frac{dy}{dx}=\frac{y\left(1-3x^2y^3\right)}{x\left(2x^2y^3+1\right)}$ such that $f(2)=1$, then $f^3(x).x^3$ is equal to

A) $x+6y$ B) $2x+6y$ C) $x^2+6y$ D) $x^3+6y^2$

3. If P= $\lim_{a\to\infty}\frac{\int_0^a \sin^6 x\cos^4 x\,dx}{a}$ and $f(x)=\frac{x^3}{2}+1-x\int_0^x g(t)\,dt$, $g(x)=x-\int_0^1 f(t)\,dt$ such that the area of the region bounded by y=f(x) with x–axis between the ordinates x=0 and x=4 is Q then the value of $\frac{64PQ}{3}$ is

A) 6 B) 2 C) 3 D) 9

4. A function $y=f(x)$ satisfies the condition $f'(x)\sin x+f(x)\cos x=1, f(x)$ being bounded when $x\to 0$. If $I=\int_0^{\pi/2} f(x)$ then which of the following is true?

A) $\frac{\pi}{2}<I<\frac{\pi^2}{4}$ B) $\frac{\pi}{4}<I<\frac{\pi^2}{2}$ C) $1<I<\frac{\pi}{2}$ D) $0<I<1$

5. If $p(x), q(x), r(x)$ and $s(x)$ are polynomials in $x$ such that $\int_1^x p(t)q(t)dt\int_1^x r(t)s(t)dt-\int_1^x p(t)s(t)dt\int_1^x q(t)r(t)dt$ is divisible by $(x-1)^{\lambda}, \lambda\in N$ and $\lambda_{\max}=a$ and $\int_b^c |\sin x|dx=8$ and $\int_0^{b+c}|\cos x|dx=9$ such that the area of the region bounded by f(x) = x sinx with x–axis and between the ordinates x=b and x=c –14b is A, then the

value of $\frac{a\sqrt{2}}{A} = \ldots\ldots$

A) $\frac{4}{\pi}$  B) $\frac{2}{\pi}$  C) $\frac{8}{\pi}$  D) $\frac{1}{\pi}$

6 . Let $f:[1,\infty)\to(-\infty,-1]$ be a continuous and differentiable function such that $f(x)=x-2x^2+\int_1^x \frac{f(t)}{x}dt$ for all $x\in[1,\infty)$. Then, which of the following statement(s) is/are CORRECT?

A) The curve y = f(x) passes through the point (3, –21)

B) Function f(x) is an onto function

C) Function f(x) is a one–one function

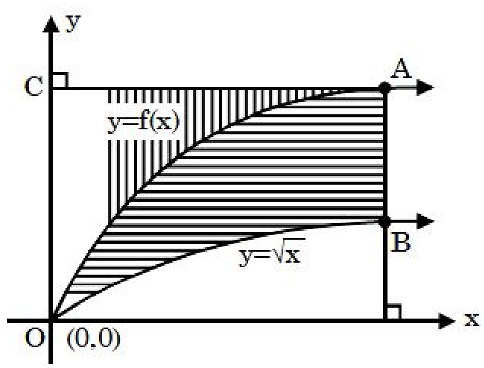
D) The area of the region $\{(x,y)\in[1,2]\times R: f(x)\le y\le -x\}$ is $\frac{5}{2}$

7. If the curve $y=f(x)$ satisfying $\int_1^x t^4 f(t)dt=\frac{x^4}{4\sqrt{3}}+3\int_x^1 f(t)dt+k,\ k\in R$, is interested by the line $y=\lambda,\ \lambda\in R$ at atleast two points, then $\lambda$ belongs to:

A) $(-1,1)$  B) $\left(-\frac{1}{4},\frac{1}{4}\right)$

C) $\left(-\frac{1}{4},\frac{1}{4}\right)-\{0\}$  D) $\left(\frac{-1}{2},\frac{1}{2}\right)$

8. Let f(x) is continuous function as shown in the figure. If the area bounded by the curve $y=f(x)$, $y=\sqrt{x}$ and line segments AB is equal to area bounded by $y=f(x)$, y–axis and line segment AC then which of the following is/are CORRECT?




A) If $\int_0^1 f(x)dx=\frac{1}{3}$ then $f\left(\frac{1}{4}\right)=\frac{1}{2}$

B) If $\int_0^1 f(x)dx=\frac{4}{3}$ then $f\left(\frac{1}{4}\right)=1$

C) There is only one such function $f$

D) There are infinitely many such function(s) $f$

9. A curve $y = f(x)$ satisfies the differential equation $\left(1+x^2\right)\frac{dy}{dx}+2yx=4x^2$ and passes through the origin. For the function $y = f(x)$ which of the following does not holds good?

A) $f$(x) has same domain and range

B) $f$(x) is strictly increasing $\forall\, x \in R$

C) $f$(x) has no inflection point

D) The area encloses by $y=f^{-1}(x)$, the x–axis and the ordinate at $x=\frac{2}{3}$ is $\frac{4}{3}\ln 2$

10. $f$(x) is defined for $x \geq 0$ & has a continuous derivative. It satisfies $f(0) = 1$, $f'(0)=0$ & $\left(1+f(x)\right)f''(x)=1+x$. The values $f(1)$ can't take is/are

A) 1.45 B) 1.75 C) 2.15 D) 1.85

11. If $f:(0,\infty)\to(0,\infty)$ for which there is a positive real number 'a' such that it satisfies differential equation $f'\left(\frac{a}{x}\right)=\frac{x}{f(x)}$, then which of the following is/are TRUE?

A) f(x) can be linear function

B) f(x) can be a functional of the type $p(x)^{1/q}; p \in R^+, q \in I^+$

C) $f'(x)$ can be positive

D) f(x) can be twice differentiable function

12. Consider f as a twice differentiable function such that $f(x)+f''(x)=-x\,g(x)\,f'(x)\ \forall x \geq 0$ where, $g(x)\geq 0\ \forall x\geq 0$, then which of the following is/are CORRECT?$(\forall\, x \geq 0)$

A) $\left(f(x)\right)^2+\left(f'(x)\right)^2$ is a non increasing function

B) $\left(f(x)\right)^2<3\left(f(0)\right)^2+\left(2f'(0)\right)^2$

C) $\left|f(x)\right|\leq\alpha$, $\alpha$ is a fixed real constant

D) $\lim\limits_{x\to\infty} f(x).\sin\left(\frac{1}{x}\right)$ exists

13. Let f(x) be a differentiable function such that $f'(x)>0, \int_0^1 f(x)\,dx = 2, \int_0^1 |f(x)|\,dx = 4$.

Choose the correct option(s).

A) f(x) = 0 has exactly one root in (0, 1)

B) f(x) = 0 has no root in (0, 1)

C) $\int_0^1 f(x).\left(\int_0^x |f(t)|\,dt\right)dx = 1$

D) $\int_0^1 f(x).\left(\int_0^x |f(t)|\,dt\right)dx = 7$

14. Consider a twice differentiable function $y = f(x)$ satisfying $f(x)+f''(x) = 2f'(x)$ where $f(0) = 0, f(1) = e$ then the value of $\int_0^1 f(x)\,dx$ is equal to

15. If A is the approximate area of the region bounded by $f(x) = e^{-x^2}$ with y–axis and the positive x–axis and $B = \int_0^\infty e^{-x^2}\cos(2020x)\,dx$ and $\sqrt{\log\left(\frac{A}{B}\right)} - 1002 = k$, then the value of k is equal to

16. If $f:[0,\infty)\to[1,\infty)$ is a quadratic function which is invertible and satisfying the equation $\left(f'(x)-4x\right)^2 + \left(f''(x)-\lambda\right)^2 = f'''(x)\ \forall\, x \geq 0$, where $\lambda$ is constant then find the value of $\frac{45}{11\lambda}\int_1^3 x f^{-1}(x)\,dx$ is

17. If the area bounded by $y = f(x), x = \frac{1}{2}, x = \frac{\sqrt{3}}{2}$ and x–axis is A sq. units where $f(x) = x + \frac{2}{3}x^3 + \frac{2}{3}\frac{4}{5}x^5 + \frac{2}{3}\frac{4}{5}\frac{6}{7}x^7 + \ldots\ldots\infty, |x|<1$; then the value of [4A] is

(Where [.] is GIF)

18. For continuous function $f:[0,1]\to R$; $A = \int_0^1 \left(x^2.f(x)\right)dx; B = \int_0^1 x.\left(f(x)\right)^2 dx$ then, maximum value of A – B is $\lambda$ where $\left[\frac{1}{4\lambda}\right]$ is______

19. If the line $x=\alpha$ divides the area of Region $S=\{(x,y)\in R^2, x^5\le y\le x^2, 0\le x\le 1\}$ into two equal parts then which of the following is/are **CORRECT**?

A) $\left(\frac{3}{4}\right)^{1/3}<\alpha<1$

B) $0<\alpha<\left(\frac{3}{4}\right)^{1/3}$

C) $\alpha=\frac{2\alpha^6+1}{4\alpha^2}$

D) $2\alpha^2+4\alpha^3-1=0$

20. A curve is given by $f(x)=x\sin\pi x$ then which of the following is/are **CORRECT**.

A) $\int_0^2 f(x)dx=-\frac{2}{\pi}$

B) $\int_0^2 f(x)dx=\frac{2}{\pi}$

C) Area between $y=x\sin\pi x$, x-axis, x = 0 and x = 2 is $\frac{4}{\pi}$.

D) Area between $y=x\sin\pi x$, x-axis, x = 0 and x = 2 is $\frac{2}{\pi}$.

21. If $y=f(x)$ defined in $(0,\pi)$ satisfies the differential equation $\frac{dy}{dx}=\sin 2x+3y\cot x$ and $y\left(\frac{\pi}{2}\right)=2$, then which of the following statement(s) is(are) **CORRECT** ?

A) $y\left(\frac{\pi}{6}\right)=0$

B) $y'\left(\frac{\pi}{3}\right)=\frac{9-3\sqrt{2}}{2}$

C) $y(x)$ increases in interval $\left(\frac{\pi}{6},\frac{\pi}{3}\right)$

D) The value of definite integral $\int_0^{\pi/2} y(x)dx$ equals $\pi$.

22. $f:\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\to R$, $y=f(x)$ has primitive F(x) such that $f(x)+\cos x.F(x)=\frac{\sin 2x}{(1+\sin x)^2}$, then which of the following is/are **TRUE**.

A) $f(x)=\frac{-2\cos x}{(1+\sin x)^2}-C\cos x e^{-\sin x}$

B) If $F(0)=2$ then $F(x)=\frac{2}{1+\sin x}$

C) If $F(0)=2$ then $F\left(\frac{\pi}{2}\right)=1$.

D) If $F(0)=2$ then $f(0)=-2$.

23. Let function $f(x)$ satisfy $x^2 f'(x) + 2xf(x) = e^x$ and $f(2) = \frac{e^2}{4}$. Which of the following is/are **CORRECT**?

A) $f(x) = 1$ has exactly one real solution.

B) $f(x) = 3$ has exactly three real solutions.

C) $f(x)$ has local maxima but no local minima.

D) $f(x)$ has local minima but no local maxima.

24. Let y(x) be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is(are) true?

(A) $y(-4) = 0$

(B) $y(-2) = 0$

(C) y(x) has a critical point in the interval $(-1, 0)$

(D) y(x) has no critical point in the interval $(-1, 0)$.

25. Let $y = f(x)$ be differentiable function and satisfy $f(0) = 2$, $f'(0) = 3$ and $f''(x) = f(x)$. Which of the following is/are **TRUE** ?

A) Range of the function y = f(x) is R

B) $f(\ell n 2) = \frac{19}{4}$

C) Area enclosed by $y = f(x)$ with co-ordinate axes in 2nd quadrant is $3 + \sqrt{2}$

D) Area enclosed by $y = f(x)$ with co-ordinate axes in 3rd quadrant is $3 - \sqrt{5}$.

26. A curve passing through (1, 1) and (0, k) satisfies the differential equation $\left(y + ye^{x/y}\right)dx + e^{x/y}(y - x)dy = 0$, then [k] is equal to (where [.] denotes greatest integer function)

27. Let 'C' be the curve passing through the point (1, 1) has the property that the perpendicular distance of the normal from origin at any point P of the curve is equal to distance of P from x-axis. If area bounded by curve 'C' and x-axis in first quadrant is $k\pi$ square units then 4k is equal to

28. If $x\frac{dy}{dx} + 2y = \log_e x$ then $\left[\frac{e^2 y(e) - y(1)}{e^2 + 1}\right]$ =______ (Where [.] is G.I.F)

29. Consider two curves $C_1 : y = \frac{1}{x}$ and $C_2 : y = \ln x$ on the xy plane. Let $D_1$ denotes the region surrounded by $C_1, C_2$ and the line x = 1 and $D_2$ denotes the region surrounded by $C_1, C_2$ and the line x = a. If $D_1 = D_2$. The value of [a] is……. where [.] is Greatest Integer Function.

30. The area enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ over the interval $\left[0, \frac{\pi}{2}\right]$ is A, value of $\left[A^2\right]$ is ________ . [ . ] is GIF

**Answer Q.31, Q.32 and Q.33 by appropriately matching the information given in the three columns of the following table.**

Column – 1: Contains information about differential equation with initial values whose particular solution is y = f(x)

Column – 2 Represent boundaries of a region, such that area bounded by it is A Sq. units

Column – 3 Value of A

| | **Column – 1** | | **Column – 2** | | **Column – 3** |
|---|---|---|---|---|---|
| I | The function y = f(x) is solution of $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ ; $x \in (-1, 1)$, satisfying $f(0) = 0$. | i | Area bounded by $y = f(x)$, $x = 0$, $x = 1$ and $y = 0$ is A | P | $A = \frac{14}{3}$ |
| II | $y = f(x)$ is solution of differential equation $ydx + y^2dy = xdy$ where $y \geq 1$ and $y(1) = 1$. | ii | Area bounded by $y = f(x)$, $x = \frac{-\sqrt{3}}{2}$, $x = \frac{\sqrt{3}}{2}$ and x-axis is A | Q | $A = 2e - 3$ |
| III | $y = f(x)$ is solution of D.E. $\frac{dy}{dx} = y + 1$ and $y(0) = 1$. | iii | Area bounded by $y = f(x)$, $x = -3$, $x = 0$ and $y = 1$ is A | R | $2\pi$ |
| IV | $y = f(x)$ is solution of D.E . $ydy + xdx = 0$, $x \geq 0$, $y \geq 0$ and $y(2) = 2$. | iv | Area bounded by $y = f(x)$, $x = 0$ and $y = 0$ is A | S | $A = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$ |

31. Which of the following options are only **CORRECT**?

A) I (ii) S B) I (iv) S C) (II) (iii) R D) II (iv) R

32. Which of the following options are only **CORRECT**?

A) II (ii) Q B) II (iii) P C) III (iii) P D) III (ii) Q

33. Which of the following options are only **CORRECT**?

A) III (i) Q B) III (iv) S C) IV (iv) R D) IV (i) Q

**Answer Q.34, Q.35 and Q.36 by appropriately matching the information given in the three columns of the following table.**

Column – 1: Contains functions in explicit forms
Column – 2: Contains linear boundaries of region, whose area to be found
Column – 3: Contains bounded area

| | **Column – 1** | | **Column – 2** | | **Column – 3** |
|---|---|---|---|---|---|
| I | $f(x)=\frac{x}{3};\ g(x)=2-x$ | i | $y=\frac{1}{4}$ | P | $\frac{1}{3}\ell n\frac{\sqrt{3}}{2}$ |
| II | $f(x)=(x+1)^2;\ g(x)=(x-1)^2$ | ii | $y=0$ | Q | 2 |
| III | $f(x)=\log_e(x+e);\ g(x)=e^{-x}$ | iii | $x=\frac{9}{4},\ y=0$ | R | $\frac{11}{32}$ |
| IV | $f(x)=\tan x;\ f(x)=\frac{2}{3}\cos x$ | iv | $x=0$ | S | $\frac{2}{3}$ |

34. Which of the following is **CORRECT** combination?
A) I (ii) R B) II (iii) P C) II (ii) P D) I (iii) R

35. Which of the following is **CORRECT** combination?
A) III (iii) Q B) II (iii) S C) II (ii) S D) III (i) S

36. Which of the following is **CORRECT** combination?
A) IV (ii) Q B) III (ii) Q C) IV (iv) R D) III (i) S

## KEY

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| B | A | D | A | C | ABCD | ABCD | ABD | CD | ABCD | ABCD | ABCD | AD | 1 | 8 |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 3 | 1 | 4 | BC | AC | AC | ABCD | ABD | AC | AB | 3 | 2 | 0 | 2 | 1 |
| 31 | 32 | 33 | 34 | 35 | 36 | | | | | | | | | |
| A | B | C | D | C | B | | | | | | | | | |

# SRI CHAITANYA ACADEMY

RAMANBHAVAN-1, VSP

**MATHEMATICS**

JSN SIR

DATE : 05-09-2023

1. If $y_1(x)$ is a solution of the differential equation $\frac{dy}{dx} - f(x)y = 0$, then a solution of the differential equation $\frac{dy}{dx} + f(x)y = r(x)$ is

A) $\frac{1}{y_1(x)}\int r(x)y_1(x)dx$  B) $y_1(x)\int \frac{r(x)}{y_1(x)}dx$

C) $\int r(x)y_1(x)dx$  D) none of these

2. The solution of the differential equation

$\left(x^2\sin^3 y - y^2\cos x\right)dx + \left(x^3\cos y\sin^2 y - 2y\sin x\right)dy = 0$ is

A) $x^3\sin^3 y = 3y^2\sin x + C$  B) $x^3\sin^3 y + 3y^2\sin x = C$

C) $x^2\sin^3 y + y^3\sin x = C$  D) $2x^2\sin y + y^2\sin x = C$

3. Area bounded by curve $y^2 = x$ and $x = 4$ is divided into 4 equal parts by the lines $x = a$ and $y = b$, then

A) $a^3 + b^3 = 8$  B) $a^3 + b^3 = 16$  C) $a^3 + b^3 = 4$  D) $a^3 + b^3 = 24$

4. Let functions $f: R \to R$ and $g: R \to R$ be defined by

$f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x) = \frac{1}{2}\left(e^{x-1} + e^{1-x}\right)$. Then the area of the region in the first quadrant bounded by the curves $y = f(x), y = g(x)$ and $x = 0$ is

A) $\left(2-\sqrt{3}\right) + \frac{1}{2}\left(e - e^{-1}\right)$  B) $\left(2+\sqrt{3}\right) + \frac{1}{2}\left(e - e^{-1}\right)$

C) $\left(2-\sqrt{3}\right) + \frac{1}{2}\left(e + e^{-1}\right)$  D) $\left(2+\sqrt{3}\right) + \frac{1}{2}\left(e + e^{-1}\right)$

**Question Stem for Question Nos. 5 and 6**

$f_1(x), f_2(x), f_3(x)$ are three polynomials of different degrees of the form of $x^n$ where $n \ge 0$. $f_1(x), f_2(x), f_3(x)$ are in G.P. $f_3(x)$ is highest degree polynomial such that $n \le 5$ and $f_1(x)$ is many to one mapping, $f_2(x)$ is bijective, $f_3(x)$ is into mapping from $R \to R$ and $f_1(1) = f_2(1) = f_3(1) = 1$. Given $f_3(x) > f_2(x) - f_1(x)$ holds for all $x \in R$.

5. Area bounded between $y=f_1(x)$ and $y=f_3(x)$ is________

6. If $\lim_{x\to 0}\left(\dfrac{\int_{f_2(x)}^{f_3(x)}\sin t\,dt}{f_1(x)\cdot f_3(x)}\right)=L$, then $\sin^{-1}(-L)=$_____

**Question Stem for Question Nos. 7 and 8:**

Let the curves $S_1: y=x^2, S_2: y=-x^2, S_3: y^2=4x-3$

7. If Area bounded by the curves $S_1, S_2, S_3$ be A then $2\tan^{-1}\left(\dfrac{1}{\sqrt{A}}\right)=$_____

8. Area bounded by the curve $S_3, y\le -1$ and the line $x=3$ is equal to____

**Question Stem for Question Nos. 9 and 10:**

Let f be a differentiable function satisfying.

$$\int_0^{f(x)} f^{-1}(t)\,dt-\int_0^x(\cos t-f(t))\,dt=0 \text{ and } f(0)=1$$

9. The number of solution(s) of the equation $\begin{vmatrix} f(2x) & f(x)\\ \sin x & 2\end{vmatrix}=0$ in $(0,2\pi)$ is _____

10. The value of $\lim_{x\to 0}\left(\left[\dfrac{\cos x}{f(x)}\right]+\left[\dfrac{\cos 2x}{f(2x)}\right]+\left[\dfrac{\cos 3x}{f(3x)}\right]+\ldots\ldots\ldots+\left[\dfrac{\cos(100x)}{f(100x)}\right]\right)$ is equal to________ [Note: where [k] denotes greatest integer less than or equal to k]

11. Area bounded by $\sqrt{x+y+1}<\sqrt{y-x+2}$ and $y<0$ is less than

A) $\dfrac{9}{16}$ B) $\dfrac{9}{5}$ C) $\dfrac{5}{4}$ D) $\dfrac{16}{5}$

12. A point P moves inside a square with vertices $A(1,1), B(-1,1), C(-1,-1)$ and $D(1,-1)$ such that minimum $\{PA, PB, PC, PD\}\le 1$. If the area of the region 'S' traced out by moving point P is $\lambda$, then which of the following is/are correct?

A) $\lambda$ is a rational number

B) $[\lambda]=3$ (Where $[\cdot]$ represents greatest integral function)

C) $\lambda$ is an irrational number

D) Probability that a randomly chosen point inside square ABCD lies in the region S is $\sec^{-1}\left(\sqrt{2}\right)$

13. Let R be the region bounded by $yx^4 - 1 + y \le 0, y \ge 0$ and $0 \le x \le 1$. If $S_R$ be the area of region R, then

A) $S_R \ge \frac{1}{8}\sqrt{\sqrt{\frac{3}{5}}} + \frac{1}{2}$

B) $S_R \le \frac{5}{8} + \frac{3}{8}\sqrt{\sqrt{\frac{3}{5}}}$

C) $S_R \ge \frac{13}{16}\sqrt{\sqrt{\frac{3}{5}}} + \frac{1}{2}\left(1 - \sqrt{\sqrt{\frac{3}{5}}}\right)$

D) $S_R \ge \frac{3}{4}$

14. Consider a function $y = f(x)$ satisfies differential equation $y''x^2 + x(y')^2 = 2xy'$ where $f(0) = 0$ and $f''(1) = 1$. Let A be area bounded by $y = f(x)$, y-axis and line $y = 2 - \frac{\pi}{2}$, then

A) $f(x)$ is strictly decreasing function

B) $f(x)$ is strictly increasing function

C) $A = 1 - \ell n 2$

D) $A = 4 - \pi$

15. Let y(x) be a solution of the differential equation $\left(x - \frac{dy}{dx}\right)\sin x = y\cos x, x \in \left(0, \frac{\pi}{2}\right]$. If $y\left(\frac{\pi}{2}\right) = 1$, then choose the correct statement(s).

A) $y(x)$ has exactly one maxima in $\left(0, \frac{\pi}{2}\right]$

B) y(x) has exactly one minima in $\left(0, \frac{\pi}{2}\right]$

C) $\lim_{x \to 0^+} y(x) = 0$

D) $0 < y\left(\frac{\pi}{4}\right) < 1$

16. Let 'C' be a curve such that the normal at any point P on it meets x-axis at A and B respectively. If BP:PA = 1:2 (internally) and the curve passes through the point (0,4) then which of the following alternatives(s) is/are correct?

A) The curves passes through $\left(\sqrt{10},-6\right)$.

B) The equation of tangent at $\left(4,4\sqrt{3}\right)$ is $2x+\sqrt{3}y=20$.

C) The differential equation for the curve is $yy'+2x=0$.

D) The curve represents a hyperbola.

17. Let an ellipse $E_1:\dfrac{x^2}{64}+\dfrac{y^2}{48}=1$ and two parabolas $P_1$ and $P_2$ both passing through ends of minor axis of $E_1$. Vertex of each of two parabolas lie on major axis of $E_1$. If $x-2y+16=0$ is common tangent of all the three curves and A be the area bounded by $P_1$ and $P_2$ then $\dfrac{A\sqrt{3}}{16}$ is

18. Let $2\left(f(x)\right)^2-\dfrac{d^2f(x)}{dx^2}f(x)+\left(\dfrac{df(x)}{dx}\right)^2=0$ and $f(0)=f(1)=-1$. Area of region bounded by $y=0, x=0, x=1$ and $y=(2x-1)f(x)$ is $2\left(\dfrac{e^{1/a}-1}{e^{1/a}}\right)$, then 'a' is?

19. Let 'C' be the curve passing through the point (1,1) has the property that the perpendicular distance of the normal from origin at any point P of the curve is equal to distance of P from x-axis. If area bounded by curve 'C' and x-axis in first quadrant satisfying $x+y-1\le 0$ is $\dfrac{\pi}{k}$ square units then k is equal to

20. A curve passes through (2,0) and the slope of tangent at P(x,y) equals $\dfrac{(x+1)^2+y-3}{x+1}$ then

A) curve is $y=x^2-2x$

B) curve is $y=x^3-8$

C) area bounded by the curve and x-axis in fourth quadrant is $\dfrac{2}{3}$ square units

D) area bounded by the curve and x-axis in fourth quadrant is $\dfrac{4}{3}$ square units

21. If differential equation is formed to the family of all the central conics centred at origin, then

A) order = 2 B) order = 3

C) degree = 1 D) degree = order = 3

22. Let C be a curve such that the normal at any point P on it meets x-axis and y-axis at A and B respectively. If BP:PA=1:2 (internally) and the curve passes through the point (0,4) then which of the following alternative(s) is/are correct?

A) The curves passes through $\left(\sqrt{10},-6\right)$

B) The equation of tangent at $\left(4,4\sqrt{3}\right)$ is $2x+\sqrt{3}y=20$

C) The differential equation for the curve is $yy'+2x=0$

D) The curve represent a hyperbola

23. Solution of the differential equation :
$\left(3\tan x+4\cot y-7\right)\sin^2 y\,dx-\left(4\tan x+7\cot y-5\right)\cos^2 x\,dy=0$ is

A) $\frac{3}{2}\cot^2 x-7\cot x+\frac{7}{2}\tan^2 y-5\tan y+4\cot x.\tan y=c$

B) $\frac{3}{2}\tan^2 x-7\tan x+\frac{7}{2}\cot^2 y-5\cot y+4\tan x.\cot y=c$

C) $3\tan^2 y-14\cot x.\tan^2 y+7\cot^2 x-10\tan y\cot^2 x+8\cot x.\tan y+2c\cot^2 x+\tan^2 y=0$

D) $3\cot^2 y-14\cot x.\cot^2 y+7\cot^2 x+10\cot y\tan^2 x+8\tan x.\cot y=0$.

24. The area bounded by the function $f(x)=|\cos x|$ and $g(x)=\sin x$ between the ordinates $x=0$ and $x=n\pi,\ n\in N$ , is equal to

A) $4n\sqrt{2}$ if n is odd B) $2n\sqrt{2}$ if n is even

C) $2\sqrt{2}(n+1)$ if n is odd D) $2\sqrt{2}(n+1)-4$ if n is odd

25. If the Area bounded by
$y=(\sin x)^{\operatorname{cosec} x}, y=(\operatorname{cosec} x)^{\sin x}, y=(\sin x)^{\sin x}$ and $y=(\operatorname{cosec} x)^{\operatorname{cosec} x}$ between the ordinates $x=0,\ x=\frac{\pi}{2}$ denoted by $A_1, A_2, A_3$ and $A_4$ then

A) $A_4$ is the greatest B) $A_1$ is the least

C) $A_2 > A_3$ D) $A_2 < A_3$

26. The solution of $x^3 \frac{dy}{dx} + 4x^2 \tan y = e^x \sec y$ satisfying $y(1) = 0$ is $\sin y = e^x (x-1) x^{-k}$ then $k =$

27. Area bounded by $2 \geq \max. \{|x-y|, |x+y|\}$ is k sq. units then k =

28. The area bounded by the curves $y = \ln x, y = ln|x|, y = |\ln x|, y = |\ln|x||$ is

29. If the area bounded by the curves $y = -x^2 + 6x - 5,\ y = -x^2 + 4x - 3$ and the line $y = 3x - 15$ is $\frac{73}{\lambda}$, then the value of $\lambda$ is

30. Let $R = \left\{x, y : x^2 + y^2 \leq 144\ and\ \sin(x+y) \geq 0\right\}$. And S be the area of region given by R, then find $S/9\pi$.

31. If the area bounded by the region [x] + [y] = n and y = k; n, $k \in N$ and $k \leq (n+1)$ and [.] is greatest integer function, in the first quadrant, is n + r, then find r.

32. $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{k^{\frac{1}{a}}\left[n^{a-\frac{1}{a}} + k^{a-\frac{1}{a}}\right]}{n^{a+1}}$ is equal to

33. A point $P(x, y)$ moves is such a way that $[x+y+1] = [x]$ (where [.] greatest integer function) and $x \in (0, 2)$. Then the area representing all the possible positions of $P$ equals

**Paragraph for Questions 51 to 52**

Let $f : R \to R$ be a differentiable function such that $f(x) = x^2 + \int_0^{\pi} e^{-t} f(x-t)\, dt$ .

34. $f(x)$ increases for

A) $x > -3$ B) $x < -2$ C) $x > -1$ D) None of the above

35. The value $\int_0^1 f(x) dx$ is

A) $\frac{1}{4}$ B) $-\frac{1}{12}$ C) $\frac{5}{12}$ D) $\frac{12}{7}$

**Paragraph for Questions 53 to 54**

Consider the curves $S_1 : y = x^2, S_2 : y = -x^2, S_3 : y^2 = 4x - 3$

36. Area bounded by the curves $S_1, S_2, S_3$ is

A) $\frac{4}{3}$sq.u B) $\frac{8}{3}$sq.u C) $\frac{1}{6}$sq.u D) $\frac{1}{3}$sq.u

37. Area bounded by the curve $S_3$, $y \leq -1$ *and the line* $x = 3$ is

A) $\frac{7}{3}$sq.u B) $\frac{11}{3}$sq.u C) $\frac{9}{2}$sq.u D) $\frac{13}{4}$sq.u

**KEY**

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| A | A | B | A | 1.33 | 0.52 | 2.09 | 2.33 | 3 | 0 | BCD | BCD | ABCD | BC | ACD |
| 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| AD | 8 | 4 | 8 | AD | BC | AD | B | BD | ABC | 4 | 8 | 4 | 6 | 8 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | | | | | | | | |
| 1 | 1 | 2 | B | C | D | A | | | | | | | | |